



# Treatment of GI Parasitic Infestations

Pharmacology GIT Block · MEDucation by Mattin Majid

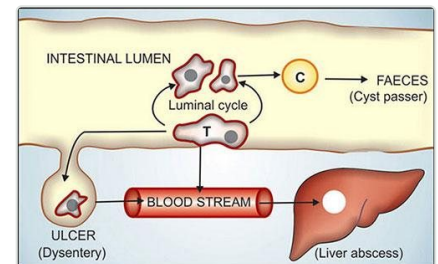
## BEFORE YOU READ — ANSWER FROM MEMORY

1. Which drug kills trophozoites but not cysts, and which drugs are luminal-only (not absorbed, or an effective luminal amebicide)?
2. Which drug class treats whipworm, pinworm, hookworm and roundworm, and what is its two-part mechanism?
3. Why is praziquantel contraindicated in ocular cysticercosis, and what should it be taken with?

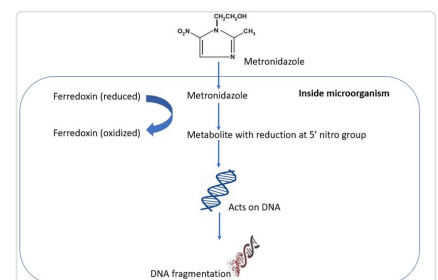
*Guessing wrong before reading still improves retention — attempt all three, then check as you go.*

## 1 Overview and amebiasis (*Entamoeba histolytica*)

|                        |                                                                                                                                                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Burden</b>          | Protozoa and helminths are among the most prevalent human infections in developing countries; more common in poverty, crowding, poor sanitation. In developed countries, <b>protozoa &gt; helminths</b> for GI disease                                                                          |
| <b>Asymptomatic</b>    | Intestinal infection with <b>cyst excretion</b> — a source for further transmission                                                                                                                                                                                                             |
| <b>Colitis</b>         | <b>Trophozoites invade the colonic mucosa</b> → colitis + bloody diarrhoea (amebic dysentery)                                                                                                                                                                                                   |
| <b>Extraintestinal</b> | Trophozoites cross the mucosa → <b>portal circulation</b> → <b>liver abscess</b> (or other sites)                                                                                                                                                                                               |
| <b>Cornerstone</b>     | <b>Metronidazole</b> or its analogue <b>tinidazole</b> — kills <b>trophozoites, not cysts</b> . Clinically effective in amebiasis, trichomoniasis, giardiasis                                                                                                                                   |
| <b>Spectrum</b>        | <b>Protozoa:</b> <i>E. histolytica</i> , <i>Giardia lamblia</i> , <i>Trichomonas vaginalis</i> .<br><b>Anaerobic bacteria:</b> <i>C. difficile</i> , <i>C. perfringens</i> , <i>H. pylori</i> , fusobacterium, <i>Bacteroides fragilis</i> , <i>Gardnerella vaginalis</i>                       |
| <b>Mechanism</b>       | <b>Prodrug:</b> nitro group reductively activated by susceptible organisms' electron-transport donors → cytotoxic reduction product → <b>acts on DNA</b> (lecture diagrams show DNA fragmentation / inhibited protein synthesis as the downstream effect). Tinidazole shares the same mechanism |



Lecture slide — luminal cycle, ulcer/dysentery, liver abscess



Lecture slides — activation and DNA-directed mechanism

| <b>Metronidazole / tinidazole</b> | <b>Details</b>                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pharmacokinetics</b>           | Readily absorbed orally; peak plasma <b>1-3 h</b> ; protein binding low (10-20%). Half-life: metronidazole <b>~8 h</b> , tinidazole <b>12-14 h</b> . Distributes widely (vaginal/seminal fluid, saliva, breast milk, CSF). Hepatic <b>CYP450 oxidation + glucuronidation</b> ; excreted mainly in urine; clearance <b>↓ in liver impairment</b>                                |
| <b>Adverse effects</b>            | Nausea, vomiting, headache, dry mouth, <b>metallic taste</b> ; rash (oral moniliasis), dysuria, dark urine; rare neurotoxicity (dizziness, vertigo, paresthesia). With alcohol: <b>disulfiram-like reaction</b> . Tinidazole: similar but <b>better tolerated</b>                                                                                                              |
| <b>Interactions</b>               | <b>Phenobarbitone, rifampin</b> (enzyme inducers) → ↓ therapeutic effect. <b>Cimetidine</b> (enzyme inhibitor) → ↓ metabolism, may need dose ↓. <b>Enhances warfarin</b> (inhibits its metabolism). <b>↓ renal elimination of lithium</b> → toxicity risk                                                                                                                      |
| <b>Clinical uses</b>              | Amebiasis, giardiasis, trichomoniasis · cutaneous leishmaniasis · sepsis/peritonitis/meningitis/brain abscess (IV) · anaerobic respiratory, intestinal, urinary infections · pseudomembranous colitis · surgical anaerobic prophylaxis · <i>H. pylori</i> peptic ulcer · perioral dermatitis, soft-tissue infection (topical) · periodontitis, ulcerative stomatitis (topical) |

## 2 Other antiamebic / antiprotozoal drugs

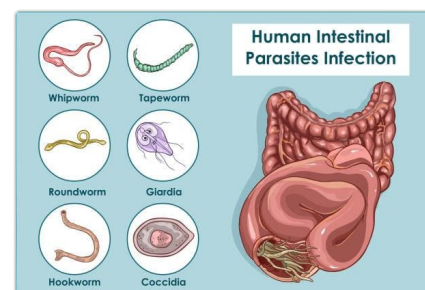
| Drug                            | Mechanism / spectrum                                                                 | Use                                                                                                                                                      | Key limitation                                                        |
|---------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Iodoquinol</b>               | Halogenated hydroxyquinoline; active against <b>luminal trophozoites and cysts</b>   | Combined with metronidazole for amebic infections                                                                                                        | <b>Not active</b> against tissue-wall or extraintestinal trophozoites |
| <b>Paromomycin</b>              | Aminoglycoside; <b>not significantly absorbed</b> from the GI tract                  | Luminal <i>E. histolytica</i> ; alternative for cryptosporidiosis and giardiasis; <b>safe choice in pregnancy</b> (avoids mutagenic risk of other drugs) | Effective <b>only</b> against luminal (intestinal) forms              |
| <b>Diloxanide furoate</b>       | Luminal amebicide                                                                    | <b>Drug of choice</b> for asymptomatic luminal infection                                                                                                 | <b>Not active</b> against tissue trophozoites                         |
| <b>Chloroquine</b>              | Eliminates trophozoites in the liver                                                 | Combined with <b>metronidazole for amebic liver abscess</b>                                                                                              | <b>Not useful</b> for luminal amebiasis                               |
| <b>Emetine / dehydroemetine</b> | Ipecac alkaloid / synthetic analogue; active against <b>tissue trophozoites</b>      | Reserve agent for tissue-invasive disease                                                                                                                | <b>Major toxicity</b> limits use                                      |
| <b>Nitazoxanide</b>             | Inhibits an enzyme essential to anaerobic energy metabolism in protozoa and bacteria | Cryptosporidiosis, giardiasis, amebiasis; <b>also active against intestinal helminths</b>                                                                | Recently approved (USA)                                               |

### Giardiasis

*Giardia intestinalis* (flagellated protozoan); ingestion of cysts from contaminated water/food. **Metronidazole 5-7 days** usually curative (may repeat/prolong). **Single-dose tinidazole** may be superior. **Paromomycin** in pregnancy

### Cryptosporidiosis

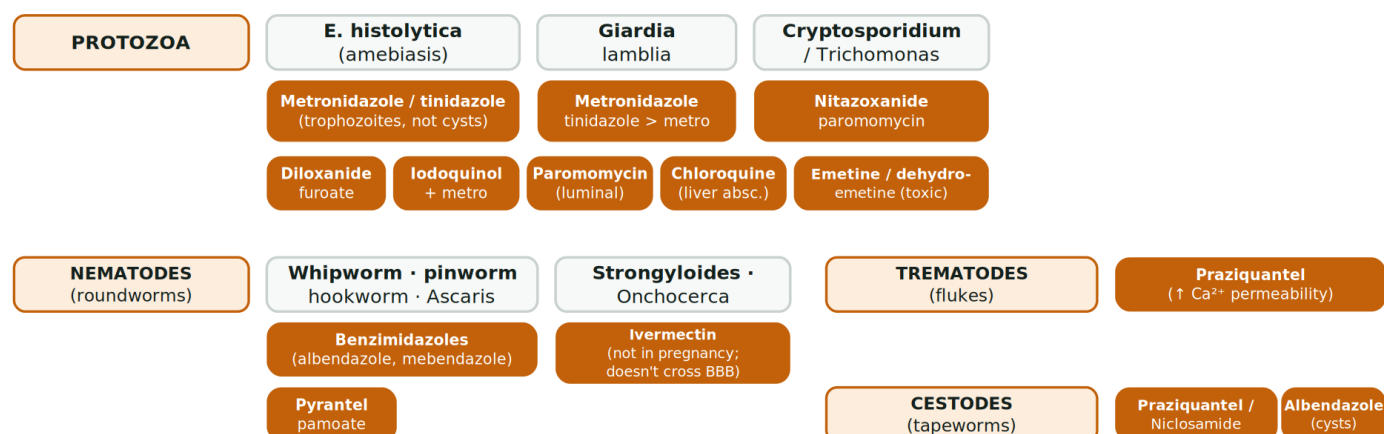
*C. parvum* / *C. hominis*; oocysts spread person-to-person or via water. Usually **self-limited**; in AIDS/immunocompromised, severity may need hospitalisation. **Nitazoxanide** active in immunocompetent patients



Lecture slide — human intestinal parasites overview

## Visual anchor — which drug treats which organism

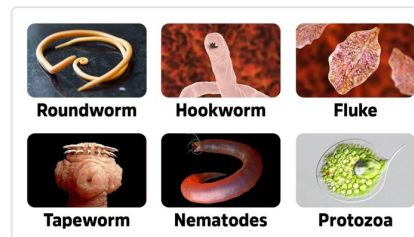
### WHICH DRUG TREATS WHICH ORGANISM



Tinted boxes are organism groups; orange tags are the drugs that act on them, positioned under the relevant organism. Discriminator to memorise: metronidazole kills trophozoites everywhere, but only diloxanide/iodoquinol/paromomycin reliably clear the luminal cyst form.

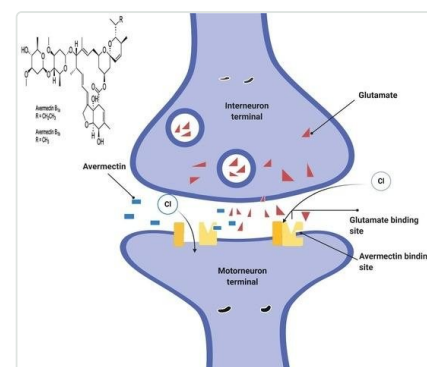
### 3 Nematodes (roundworms)

|                        |                                                                                                                                                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Benzimidazoles</b>  | Only <b>albendazole</b> and <b>mebendazole</b> in wide use, of the many synthesised. Bind parasite <b><math>\beta</math>-tubulin</b> , inhibit its polymerisation (microtubule synthesis); <b>irreversibly block glucose uptake</b> → parasite expelled in feces |
| <b>Coverage</b>        | <b>First-line</b> for whipworm ( <i>Trichuris trichiura</i> ), pinworm ( <i>Enterobius vermicularis</i> ), hookworm ( <i>Necator americanus</i> , <i>Ancylostoma duodenale</i> ), roundworm ( <i>Ascaris lumbricoides</i> )                                      |
| <b>Adverse effects</b> | Abdominal pain, diarrhoea; <b>avoid in pregnancy</b>                                                                                                                                                                                                             |



Lecture slide — albendazole MOA flowchart

| Drug                     | Mechanism                                                                                                                                                                       | Coverage                                                                                                                                                                                                     | Notes                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Pyrantel pamoate</b>  | <b>Depolarising neuromuscular blocker:</b> releases ACh, inhibits cholinesterase → paralysis; poorly absorbed orally, acts in the gut                                           | Roundworms, pinworms, hookworms                                                                                                                                                                              | Worm loses its hold on the gut wall and is expelled                     |
| <b>Ivermectin</b>        | Targets <b>glutamate-gated <math>\text{Cl}^-</math> channels</b> in nematode nerve/muscle → hyperpolarisation → paralysis and death                                             | <b>Drug of choice:</b> cutaneous larva migrans, strongyloidiasis, onchocerciasis (kills microfilariae, <b>not adult worms</b> ). More effective in <b>ascariasis/enterobiasis</b> than trichuriasis/hookworm | Oral; does <b>not readily cross the BBB</b> ; <b>avoid in pregnancy</b> |
| <b>Ivermectin target</b> | Binds glutamate-activated $\text{Cl}^-$ channels specific to invertebrate nerve/muscle synapses — increases intracellular $\text{Cl}^-$ → hyperpolarisation → flaccid paralysis |                                                                                                                                                                                                              |                                                                         |



Lecture slide — ivermectin binding at the glutamate-gated  $\text{Cl}^-$  channel

### 4 Trematodes (flukes) and cestodes (tapeworms)

|                   |                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trematodes</b> | Leaf-shaped flatworms — blood, liver, lung or intestinal flukes. <i>Schistosoma</i> , <i>Fasciola</i> , <i>Clonorchis</i> , <i>Opisthorchis</i> , <i>Paragonimus</i>         |
| <b>Cestodes</b>   | Flat, segmented, attach to host intestine: <i>Taenia saginata</i> (beef), <i>Taenia solium</i> (pork), <i>Hymenolepis nana</i> (dwarf), <i>Diphyllobothrium latum</i> (fish) |

| Drug                | Mechanism                                                                                                                          | Use                                                                                                                                   | Key point                                                                                                         |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Praziquantel</b> | ↑ <b>membrane permeability to <math>\text{Ca}^{2+}</math></b> in trematode/cestode cell membranes → paralysis, dislodgement, death | All schistosome species; most other trematode and cestode infections                                                                  | <b>Contraindicated in ocular cysticercosis</b> — destroying the organism in the eye may cause irreversible damage |
| <b>Albendazole</b>  | Same benzimidazole mechanism ( $\beta$ -tubulin, glucose uptake), effective against most nematodes                                 | <b>Primary use: cestodal disease</b> — cysticercosis, hydatid disease; also effective in <b>microsporidiosis</b> (a fungal infection) | Nematode activity is secondary to its cestode role in this lecture's emphasis                                     |
| <b>Niclosamide</b>  | Not detailed beyond its role as an alternative agent                                                                               | <b>Alternative to praziquantel</b> for cestode infections (taeniasis, diphyllobothriasis, others)                                     | —                                                                                                                 |

| Praziquantel — pharmacokinetics | Details                                                                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Absorption</b>               | Rapid; ~80% <b>bioavailable</b> orally. Take <b>with food</b> (not chewed — bitter taste)                                                    |
| <b>Distribution</b>             | Peak serum <b>1-3 h</b> ; distributes throughout the body, highest in liver/kidneys; reaches <b>CSF</b> ; ~80% <b>protein-bound</b>          |
| <b>Adverse effects</b>          | Dizziness, malaise, headache, GI upset                                                                                                       |
| <b>Interactions</b>             | Bioavailability ↓ <b>by CYP inducers</b> (phenytoin, carbamazepine, phenobarbital) or corticosteroids; plasma level ↑ <b>with cimetidine</b> |

## P Must-remember pearls

- **Metronidazole/tinidazole** kill trophozoites, not cysts — cornerstone for amebiasis, giardiasis, trichomoniasis; watch for the **disulfiram reaction** with alcohol and the **cimetidine / rifampin / warfarin / lithium** interactions.
- **Luminal-only amebicides**: diloxanide furoate (drug of choice, asymptomatic), paromomycin (not absorbed, safe in pregnancy), iodoquinol (luminal trophs + cysts) — none of these clear tissue disease.
- **Chloroquine** pairs with metronidazole for liver abscess; **emetine/dehydroemetine** hit tissue trophozoites but are limited by toxicity; **nitazoxanide** is the broadest (crypto, giardia, amebiasis, helminths).
- **Benzimidazoles** (albendazole, mebendazole):  $\beta$ -tubulin + glucose-uptake block — first-line for whipworm/pinworm/hookworm/roundworm; **albendazole's primary role is cestodal** (cysticercosis, hydatid).
- **Ivermectin** = glutamate-gated  $\text{Cl}^-$  channels, drug of choice for onchocerciasis/strongyloidiasis, no BBB crossing, avoid in pregnancy. **Praziquantel** = ↑  $\text{Ca}^{2+}$  permeability, covers schistosomes + most cestodes/trematodes, contraindicated in ocular cysticercosis.

## Q Test yourself

**1. A pregnant woman has amebic liver abscess. Metronidazole is combined with which drug, which clears hepatic trophozoites but is not useful for luminal amebiasis?**

- A.** Chloroquine   **B.** Iodoquinol   **C.** Paromomycin   **D.** Nitazoxanide   **E.** Pyrantel pamoate

**2. A patient with luminal-only amebiasis (asymptomatic cyst passer) needs treatment. Which drug is the drug of choice, since it is an effective luminal amebicide but not active against tissue trophozoites?**

- A.** Metronidazole   **B.** Diloxanide furoate   **C.** Chloroquine   **D.** Emetine   **E.** Praziquantel

**3. A patient with a nematode infection is given a benzimidazole. Which mechanism explains how it kills the worm?**

- A.** It increases membrane permeability to calcium, causing paralysis  
**C.** It binds parasite  $\beta$ -tubulin, blocking microtubule polymerization, and irreversibly blocks the parasite's glucose uptake  
**E.** It coats the worm's cuticle, preventing it from gripping the intestinal wall  
**B.** It opens glutamate-gated chloride channels, hyperpolarizing the worm  
**D.** It is reduced by nitroreductase to a cytotoxic metabolite that binds DNA

**4. A patient with schistosomiasis is treated with praziquantel. Which statement about its use is correct?**

- A.** It is the first-line agent for all nematode infections, including ascariasis  
**C.** It is ineffective against every cestode infection  
**E.** Its bioavailability increases when taken with phenytoin or carbamazepine  
**B.** It should be taken on an empty stomach, since food destroys it before absorption  
**D.** It is contraindicated for ocular cysticercosis, because destroying the organism in the eye can cause irreversible damage

**5. A patient with onchocerciasis (river blindness) is treated with ivermectin. Why does the lecture say ivermectin carries little risk of CNS toxicity despite paralyzing the worm's nerve and muscle cells?**

- A.** It is inactivated by hepatic CYP450 before it can reach the CNS  
**C.** It kills only adult worms, which do not migrate to the CNS  
**E.** It does not readily cross the blood-brain barrier  
**B.** It only affects glutamate receptors that exist in worms, not in humans  
**D.** It is not absorbed from the gastrointestinal tract at all

#### ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Chloroquine is combined with metronidazole for amebic liver abscess; it clears hepatic trophozoites but not luminal disease.
- 2 — B** — Diloxanide furoate is an effective luminal amebicide, inactive against tissue trophozoites — drug of choice for asymptomatic luminal infection.
- 3 — C** — Benzimidazoles bind  $\beta$ -tubulin and block its polymerization, and irreversibly block glucose uptake — the worm is then expelled in the feces.
- 4 — D** — Praziquantel is contraindicated in ocular cysticercosis for this reason; it should in fact be taken WITH food, and CYP inducers REDUCE (not increase) its bioavailability.
- 5 — E** — Ivermectin acts on glutamate-gated  $\text{Cl}^-$  channels in nematode nerve/muscle; it is given orally and does not readily cross the blood-brain barrier.

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#### SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Begard O. Berzinji lecture slides, LG 1 (full deck)  
Study alongside the lecture, not instead of it.